JUAN PABLO ROJO

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Cordoba, Argentina — Open to remote worldwide

PROFESSIONAL SUMMARY

Biomedical Engineer (UNC, 2026) specializing in AI for medical imaging. Built TraumaVision AI — a clinical decision support system achieving AUC-ROC 0.973 and 97.2% accuracy in fracture detection, comparable to expert radiologists (Lindsey et al., PNAS 2018). Founder of Eleven Estudio Digital, delivering AI-powered web solutions for SMEs. End-to-end experience across deep learning, full-stack web development, and clinical biomedical engineering.

KEY PROJECTS

TraumaVision AI — Clinical Decision Support System 2025 – 2026

Bachelor's Thesis | Universidad Nacional de Cordoba

- Built an AI web application to assist traumatologists in detecting fractures from X-ray images.
- Trained DenseNet classifier: AUC-ROC 0.973 | Accuracy 97.18% | Sensitivity 90.85% | Specificity 98.64% | F1 92.36% on a blind test set (n=817, FracAtlas dataset).
- Integrated Grad-CAM explainability to generate heatmaps highlighting fracture regions for clinical staff.
- Trained YOLOv8m for fracture localization on pediatric wrist X-rays (GRAZPEDWRI dataset).
- Built full-stack web app: FastAPI, SQLAlchemy, SQLite, Google OAuth, PDF report generation (ReportLab), interactive analytics dashboard.
- Identified and corrected Clever Hans shortcut learning via targeted augmentation (RandomErasing, RandomAffine, RandomResizedCrop).
- Stack: Python | PyTorch | DenseNet | YOLOv8 | Grad-CAM (Captum) | FastAPI | HTML/CSS/JS

TECHNICAL SKILLS

AI & Machine Learning: PyTorch, Scikit-learn, NumPy, Pandas | transfer learning, fine-tuning | Claude API | medical image processing

Computer Vision: DenseNet, YOLOv8 (Ultralytics), Grad-CAM (Captum) | image classification, object detection, clinical image analysis

Web Development: Next.js, React, FastAPI, SQLAlchemy, SQLite, Supabase | HTML/CSS/JS | GSAP | Vercel

Data & Visualization: Matplotlib, Pandas | interactive dashboards | Excel (advanced)

Biomedical Engineering: Corrective & preventive medical equipment maintenance | Joint Commission standards | ISO/IEC/ANMAT | SaMD/CDSS

Tools & Marketing: Python, Git/GitHub, Vercel | Meta Ads, Google Ads, SEO

WORK EXPERIENCE

Founder — Web Development & AI Agents 2025 – present

Eleven Estudio Digital | Cordoba, Argentina | eleven-pagina.vercel.app

- Founded and lead a digital studio delivering web development, AI agents, automation, and digital content for SMEs and entrepreneurs in Argentina.
- Built client websites with Next.js, React, GSAP, and Vercel; integrated AI agents via Claude API for 24/7 automated customer service on WhatsApp and Instagram.
- Managed the full project lifecycle: discovery, strategy, design, development, delivery, and ongoing monthly maintenance.
- Ran Meta Ads and Google Ads campaigns and ongoing SEO strategies for clients in the services and retail sectors.

Biomedical Engineering Intern Apr – Jul 2024

Sanatorio Allende S.A. | Nueva Cordoba, Argentina

- Performed corrective and preventive maintenance on medical equipment across operating rooms, adult & pediatric ICU, inpatient wards, and sterilization units.
- Coordinated with medical and technical staff to diagnose failures and restore critical device availability.
- Ensured compliance with Joint Commission International (JCI) standards across all maintained equipment and clinical processes.

Undergraduate Teaching Assistant — Human Physiology Aug 2023 – Jun 2024

Universidad Nacional de Cordoba, FCEFyN | Cordoba, Argentina

- Supported instruction of Human Physiology for undergraduate engineering students, translating complex concepts into accessible content.
- Facilitated collaborative learning sessions and coordinated study groups of 15+ students.

Rentals Technician Dec 2023 – Mar 2024 | Dec 2024 – Mar 2025

Snow Summit LLC | Big Bear Lake, CA, USA

- Developed professional-level English fluency in a fully English-speaking, high-volume technical environment.
- Managed equipment rental processes and performed routine technical maintenance.

EDUCATION

Biomedical Engineering (B.Eng.) 2020 – 2026 (expected)

Universidad Nacional de Cordoba — Faculty of Exact, Physical and Natural Sciences (FCEFyN)

GPA: 7.45/10 (no failures) | 90% degree completion

Thesis: TraumaVision AI: A Clinical Decision Support System for Fracture Detection Using Deep Learning

ADDITIONAL TRAINING

Applied Data Science with Python Specialization Coursera

University of Michigan

Data visualization, applied machine learning, network analysis, supervised learning. Libraries: NumPy, Pandas, Matplotlib, Scikit-learn.

LANGUAGES

English: B2 Upper-Intermediate — EF International Language Campus, Eastbourne, UK + 2 seasons of full-time professional work in California, USA

Spanish: Native